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Adolescent Executive Function as a Resilience Factor in the Family Stress Model Among Mexican-Origin Families

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Abstract

Mexican-origin families often face economic hardship due to systemic oppression, increasing the likelihood of adolescent marijuana use. While the Family Stress Model provides insight into the mechanism of the association between family economic hardship and adolescent marijuana use, resilience factors are relatively unknown. The current study uses a three-wave longitudinal dataset of Mexican immigrant families in the United States (Wave 1: Adolescents: M_{age}= 12.29, *SD* = 0.93; Mothers: M_{age}= 38.57, *SD* = 5.71) to investigate the mechanism underlying the association between family income and adolescents' likelihood of marijuana use, guided by the Family Stress Model. The protective role of adolescent executive function (shifting task performance and working memory), which have been widely linked to adolescent marijuana use in prior research, was tested as a key resilience factor supporting behavioral regulation and adaptive coping in negative family environments. The results revealed the long-term detrimental influence of early adolescents' family income on the likelihood of using marijuana in late adolescence through family economic pressure, maternal internalizing symptoms, maternal hostility toward partner, and maternal hostility toward adolescent. The downstream link is buffered by adolescent's longer reaction times in shifting tasks and longer digit span by attenuating the influence of maternal hostility toward adolescent on the likelihood of adolescent marijuana use. Revealing the mechanism and identifying resilience factors for the association between family economic hardship and adolescent's marijuana use in Mexican-origin families shed light on the targets of interventions to help adolescents thrive and overcome economic disadvantage in Mexican-origin communities.

Introduction

Despite shifting cultural and legal norms surrounding marijuana use, marijuana use is linked to numerous adverse outcomes to even recreational users, such as psychiatric disorders and poor academic achievement (Sultan et al., 2023). Latino adolescents display elevated risk of marijuana use relative to their non-Hispanic White peers (Goodwin & Silverman, 2024). As Mexican-origin individuals account for the largest proportion of Latinos (Moslimani et al., 2023), it is essential to examine processes which underly their marijuana use in order to improve their health and reduce racial/ethnic disparities in substance use. The current study innovates by examining Mexican-origin youth's marijuana use as an outcome of processes delineated by the Family Stress Model (FSM, Masarik & Conger, 2017), which links youth's adjustment to familial functioning within the context of economic hardship. Of equal importance is the examination of adolescents' individual-level resiliency factors that can mitigate the downstream effects of family economic stress. Executive function is well-documented as a critical predictor of adolescent substance use, including marijuana use (Cavalli et al., 2023; Coelho et al., 2024). However, it remains unclear how executive function may serve as a resilience factor that buffer the downstream negative influence of family economic stress on Mexican-origin adolescents' marijuana use. By studying the moderating effect of executive function within the family stress model framework, the current study offers an important contribution to the literature and provides meaningful insights for designing marijuana use prevention and intervention programs tailored to the needs of Mexican-origin youth.

Family Stress Model in Mexican-Origin Families

The Family Stress Model (FSM) offers a developmental framework that links economic disadvantage to a cascade of family stress processes, including economic pressure, parental depression and anxiety, hostility toward a partner, and hostility toward children. This process ultimately jeopardizes children's positive development (Masarik & Conger, 2017). Understanding the FSM within the context of Mexican-origin families is particularly important, given the disproportionate economic and systemic inequities faced by this population. Mexican-origin communities display lower household incomes than the national median (Moslimani et al., 2023), which can be attributed to structural barriers to educational attainment and employment opportunities which impact Latino communities more broadly (Marrero, 2016).

Moreover, substance use among Mexican-origin adolescents is also a pressing concern (Goodwin & Silverman, 2024). To address this issue, a growing body of research has examined how family processes influence substance use among Latino youth. For example, diminished parental involvement has been linked to lifetime marijuana and cigarette use (Zapata Roblyer et al., 2016), family conflict has been shown to predict the trajectories of adolescent alcohol use and binge drinking (Martinez et al., 2025), and parental monitoring has been identified as a protective factor against substance use, including marijuana use, among Mexican-origin youth (Atherton et al., 2016). However, when parenting and parent-child interactions are accounted for, family economic stress and socioeconomic status do not consistently predict substance use (Atherton et al., 2016; Zapata Roblyer et al., 2016). This pattern suggests that the influence of distal factors (e.g., family economic conditions) on adolescent substance use may be mediated by more proximal family processes (e.g., parenting), highlighting the need to examine the cascading

pathways proposed in the FSM.

Past work has substantiated the applicability of the overall FSM to Latino adolescents' adjustment, such as emotional and social adjustment (Miller & Csizmadia, 2022; Taylor et al., 2024), including studies that specifically draw on Mexican-origin adolescent samples to examine how family processes influence adolescent internalizing and externalizing problems, and academic performance (Delgado et al., 2013; White et al., 2015). One study extended the FSM to substance use behaviors among Mexican-origin youth, demonstrating how economic hardship contributes to parent distress, interparental conflict, and poor parenting, which further increase youth's eventual substance use (i.e., alcohol, cigarette, and marijuana use,Martin et al., 2019). Importantly, although the family stress model has elucidated how economic hardship cascades into adolescent maladjustment (e.g., marijuana use), its application to Mexican-origin populations has predominantly framed youth as passive recipients of stressors, neglecting youth's individual-level resilience and adaptative mechanisms. Integrating adolescents' executive function into the family stress model addresses this gap by theorizing how adolescents actively mitigate the impact of family environment changes following economic stress. Executive function may buffer the negative effects of family stress by weakening the influence of maternal hostility on adolescent marijuana use, as it supports adolescents' ability to regulate behavior by shifting attention to manage emotions and consider adaptive alternatives (Grenard et al., 2008), to ultimately reduce their reliance on maladaptive coping strategies such as substance use.

Resilience Factors within the Family Stress Model in Mexican-origin Families

The FSM conceptualizes the family as an interdependent system, where economic strain

relates to poor parental mental health, intimate partner relationships, and parenting (Masarik & Conger, 2017). Salient protective mechanisms can lie within youth along the prospective pathway from economic hardship during early adolescence to marijuana use during late adolescence (the outcome of interest in present study), which research has not fully explored among Mexican-origin populations. As a robust predictor of lower marijuana use (Cavalli et al., 2023), executive function may also actively disrupt risk pathways based on the family stress model, buffering the impact of maternal hostility toward adolescent on youth marijuana use specifically. Investigating the buffering role of executive function is critical in understanding how youth recalibrate maladaptive stress responses even when structural inequities limit parents' capacity to mitigate hardship.

Executive function is a set of deliberate "top-down" cognitive processes that enable the regulated processing of information and actions to support goal-directed behavior (Mason et al., 2024). Executive function plays a crucial role in regulating and modulating approach tendencies triggered by impulsive and intuitive stimuli. In the context of heightened maternal hostility—stemming from economic pressure, poorer maternal mental health, and subsequent intimate partner hostility—adolescents' executive function may serve as a resilience factor against marijuana use. Specifically, stress resulting from maternal hostility may activate impulsive reactions, increasing reward-driven behaviors such as marijuana use. In such situations, executive function may help adolescents regulate their impulsive tendencies by focusing on task-relevant information (baseline attention in the shifting task) and by redirecting attention away from stress-inducing contexts (reorient attention in the shifting task), and by using working

memory to actively maintain goals and alternative behavioral options in mind, even in the presence of family stressors like maternal hostility (Grenard et al., 2008).

One study on Mexican-origin youth found that effortful control can help mitigate the negative effects of economic hardship on youth active coping, supporting the conceptualization of executive function as a self-regulatory resilience factor in the context of family stress and adolescent marijuana use (Taylor et al., 2018). A recent longitudinal study also found that better executive function can buffer the negative effects of harsh parenting on children's externalizing problems (Hong et al., 2024). However, they used principal components analyses that combine multiple executive function domains, including shifting task performance and working memory (Hong et al., 2024). This approach makes it difficult to differentiate the unique contributions of each subcomponent. Although working memory and shifting task performance are closely intertwined (Draheim et al., 2016), prior research suggests that these components may exert distinct influences on adolescent marijuana use (Guo et al., 2023). Therefore, distinguishing between working memory and shifting task performance is essential for understanding which aspects of executive function can protect adolescents from engaging in marijuana use in the context of maternal hostility.

Working memory refers to the ability to temporarily hold and manipulate information (Grenard et al., 2008). Working memory is essential for maintaining multiple pieces of information, such as goals and behavioral options, in an active state, particularly in the presence of contextual stressors that may trigger reward-seeking behaviors like marijuana use (Grenard et al., 2008). Deficits in working memory are significantly related to substance use (Khemiri et al.,

2019; Khurana et al., 2013). In the context of maternal hostility, stronger working memory may allow adolescents to retain and consider multiple coping strategies, helping them manage maternal hostility in a more adaptive way rather than using marijuana. Additionally, stronger working memory may help adolescents keep their goals and aspirations, such as completing their education, pursuing a desired career, and maintaining future health in an active state, even in emotionally charged situations. By helping adolescents stay focused on their long-term goals, working memory can support their ability to resist the temptation of immediate relief through marijuana use.

Shifting task performance includes both sustaining attention on a set of instructions or stimuli over time and reorienting attention in response to new or changing information (Gresch et al., 2024). Better performance in shifting task is directly linked to reduction in substance use during adolescence (Gustavson et al., 2017). In the face of adversity, a strong ability to focus on task-relevant information (i.e., baseline attention) and to redirect attention away from stressful cues (i.e., reoriented attention), may serve as forms of individual-level resilience. These attentional skills can enhance adolescents' well-being by supporting mood regulation and helping them cope with stress (Davidovich et al., 2016). Such attentional skills allow them to stay focused on their own goals during family conflict and to flexibly shift attention away from, and become less reactive to, hostile behavior or negative cues from their mother (Mason et al., 2024).

Prior studies often focus on reaction time in shifting tasks, with shorter reaction times being interpreted as indicative of better performance (Draheim et al., 2019). However, this approach overlooks the trade-off effects between reaction time and accuracy (Draheim et al., 2016).

Adolescents with longer reaction times may engage in a more cautious decision-making process, slowing down to ensure accuracy (Chevalier et al., 2013). Given this potential trade-off effect, the present study examines reaction time in the shifting task as a moderator between maternal hostility toward adolescent and marijuana use while including accuracy as a covariate. Because accuracy is often interpreted as another indicator of better performance, the first sensitivity analysis tests the moderating effect of accuracy while controlling for reaction time. Additionally, because the protective role of reaction time in the context of maternal hostility may vary depending on adolescents' accuracy levels, a second sensitivity analysis tests a three-way interaction among accuracy, reaction time, and maternal hostility toward adolescent in predicting adolescent marijuana use.

Current Study

The current study draws upon and extends the FSM through two primary objectives (see Figure S1). First, as guided by the FSM, the present study examines how familial economic hardship (i.e., family income) and economic pressure during early adolescence (Wave 1) initiate a cascade of negative familial outcomes (i.e., maternal internalizing symptoms, maternal hostility toward partner) by middle adolescence (Wave 2), which then spill over to maternal hostility toward adolescent, ultimately contributing to poorer youth outcomes, such as marijuana use, in late adolescence (Wave 3) (b1 -> b2 -> b3 -> b4 -> b5, Figure S1). Second, the study tests the potential protective roles (via tests of moderated mediation) of adolescents' executive function at Wave 3. More specifically, two types of executive function, working memory and shifting task performance (including baseline attention and re-oriented attention), are examined as separate

moderators (c1's influence on b5, Figure S1) of the relationship between maternal hostility toward adolescent and adolescent marijuana use, and conditional indirect effects are tested to explore how the family stress process, originating from Wave 1 family income and culminating in Wave 3 adolescent marijuana use (b1 → b2 → b3 → b4 → b5, Figure S1), varies depending on levels of each executive function components. The hypothesis is that better working memory and shifting task performance (in both baseline attention and re-oriented attention) will mitigate the negative effects of maternal hostility toward adolescent on adolescents' marijuana use.

Methods

We state below how we determined our sample size, all data exclusions, all manipulations, and all measures in the study.

Participants

The data used in this study is from a three-wave longitudinal study spanning 2012 and 2020 (Wave 1: 2012–2015; Wave 2: 2013–2016; Wave 3: 2017–2020). A priori power analysis was not conducted at the time of initial data collection and all available sample was utilized given attrition across the longitudinal study. The study exclusively utilized families that participated in Wave 3, as the research concentrated on adolescent marijuana use, which was only collected in Wave 3. As a result, the study included 334 Mexican-origin adolescents and 329 of their mothers from low-income immigrant households in central Texas, USA. Of relevance to the study sample over the course of the study period is the legal context of marijuana in Texas, as medical marijuana was legalized in 2015 (during Wave 2) while recreational marijuana has remained illegal (Texas State Law Library, 2023). At Wave 1, the average family income ranged from $20,001 to $30,000, with

the parental average education level being middle or junior high school level. Almost all families (99%) in the study were two-parent families at Wave 1. At Wave 1, adolescents were in early adolescence (N_1 = 604, M_{1age} = 12.29, SD = 0.93), Wave 2 occurred during middle adolescence (N_2 = 483, M_{2age} = 13.12, SD = 0.94), and Wave 3 data was captured during late adolescence (N_3 = 334, M_{3age} = 17.10, SD = 1.12). The age of mothers ranged from 27 to 61 years old (M_{1age} = 38.57, SD = 5.71; M_{2age} = 39.37, SD = 6.04; M_{3age} = 43.33, SD = 5.75). Slightly over half of the adolescent participants identified as female (N = 183, 55.3%). About 76% of the adolescents were U.S. born (N = 254), 24% of adolescents were born in Mexico, and 99% of the mothers were born in Mexico (N = 327), while the rest of the mothers were U.S.- born. Attrition analysis indicates that adolescents whose mothers had higher levels of education at Wave 1 were more likely to remain in Wave 2 ($t(591) = -2.41$, $p < .05$). Additionally, younger adolescents and those with mothers with higher educational levels at Wave 1 (age: $t(602) = 3.49$, $p < .001$; maternal education: $t(591) = -3.45$, $p < .001$) and Wave 2 (age: $t(480) = 3.20$, $p < .01$; maternal education: $t(476) = -3.39$, $p < .001$) were more likely to continue participation at Wave 3.

Procedure

Participants' families were initially recruited through school presentations, community recruitment, and public records. Families were selected if parents were of Mexican heritage and their children used English and Spanish to translate for at least one parent. Informed consent from parents and informed assent from adolescents were obtained. Family visits occurred in all three waves, where research assistants read the questionnaires to mothers and adolescents and recorded their responses on a laptop. At Wave 3, adolescents also completed executive function tasks. The

tasks were administered before the questionnaire session. All materials were provided in both English and Spanish. The questionnaires were originally crafted in English and translated to Spanish by bilingual and bicultural research assistants. Families received $60 in compensation at Wave 1, and $90 at Waves 2 and Wave 3 for their participation. All procedures were approved by the Institutional Review Board at the University of Texas at Austin (Protocol Number: 2015-01-0006, Study Title: Collaborative Research: Mexican American Language Brokers' Multiple Levels of Stress and Academic and Health Outcomes). The study was not preregistered. Data and analysis codes are available at https://osf.io/ezkgu/ (Kim et al. 2025).

Measures

Family Income

At Wave 1, mothers reported family income on a scale ranging from 1 (*$10,000 or under*) to 11 (*$110,001 or more*) by answering, "Think of all the income from persons in your family who live in the same house with you. Consider all kinds of income including wages earned from jobs, alimony, unemployment compensation, and government assistance. What was your family's income from all sources before taxes during the past year?"

Family Economic Pressure

At Wave 1, mothers answered nine yes (1) or no (0) questions from the Financial Cutbacks scale (Matthews et al., 1996), regarding whether the family had made certain financial adjustments based on financial need in the past three months. A sample item is "Has your family (or a member of your family) changed food shopping or eating habits a lot to save money." Items were summed and higher values represent higher levels of family economic pressure.

Maternal Internalizing Symptoms

At Wave 2, maternal internalizing symptoms were assessed using a latent factor score of mother-reported anxiety and depressive symptoms. Anxiety was measured using four items from the Generalized Anxiety Disorder-7 and Manifest Anxiety Scale (Reynolds & Richmond, 1997; Spitzer et al., 2006)), rated from 1 (*not at all*) to 4 (*nearly every day*). A sample item is "How often have you been feeling nervous, anxious or on edge?" (α = .85). Depressive symptoms were measured using 20 items from the Center for Epidemiologic Studies of Depression Scale (Radloff, 1977), rated from 1 (*rarely or none of the time*) to 4 (*most or all of the time*). A sample item is "I thought my life had been a failure." (α = .89). The CES-D includes four subscales (αs = .66 -.90): depressed affect, positive affect, somatic complaints/activity inhibition, and interpersonal difficulties. A latent variable was created using the four anxiety items and the mean scores of CES-D subscales to generate internalizing factor score. Details of the conceptual model and fit indices are presented in supplementary confirmatory factor analyses and Figure S2.

Maternal Hostility toward Partner

At Wave 2, mothers reported their hostility toward partners (adolescents' fathers) using six items adapted from the Iowa Youth and Families Project on a scale ranging from 1 (*never*) to 7 (*always*) (Matthews et al., 1996). A sample item is "How often do you get angry at your partner?" Higher mean values represent higher maternal hostility toward their partners (α = .85).

Maternal Hostility toward Adolescent

At Wave 3, mothers reported on their hostility toward adolescent using six items from the Iowa Youth and Families Project, ranging from 1 (*never*) to 7 (*always*) (Flynn et al., 2018). A

sample item is "How often do you shout or yell at your child because you were mad?". Higher mean scores represent higher levels of maternal hostility toward adolescent (α = .88).

Adolescents' Lifetime Marijuana Use

At Wave 3, adolescents reported on their lifetime marijuana use by answering one question, "During your lifetime, how many times have you used marijuana (pot, weed)?" ranging from 0 (*0 times*) to 6 (*100 or more times*) (Kim et al., 2021). Because 65.4% of adolescents reported no lifetime marijuana use, and only 34.6% reported any use (i.e., 9.5% for 1-2 times, 10.1% for 3-9 times, 4.6% for 10-19 times, 4.6% for 20-39 times, 3.4% for 40-99 times, and 2.4% for 100 or more times), the variable was recoded into a dichotomous variable (0 = *no use*, 1 = *any use*).

Adolescent Executive Function

At Wave 3, adolescents' executive function was measured through working memory (Backward Digits Span task) and shifting task (Color-Shape Task). Participants recall digits in reverse order of their presentation, with higher scores indicating stronger working memory. Shifting task consists of stay (*baseline* attention) and switch (*reorienting* their attention) trials in which participants categorize images according to their color or shape, as quickly and accurately as possible. See supplemental methods for a detailed description of executive function tasks.

Covariates

This study's covariates (see supplemental covariates section) included demographic variables assessed at Wave 1, including adolescents' binary gender, age, nativity, and maternal educational level, which were found to be associated with adolescent marijuana use in previous research (Aschengrau et al., 2021; Fattore & Fratta, 2010; Schramm-Sapyta et al., 2009; Warner et al.,

2008).

Analysis Plan

Logit regression models were estimated using structural equation modeling in Mplus 8.3. Maximum likelihood estimation with robust standard errors (MLR) was used for all models to address potential non-normality of variables, and full information maximum likelihood (FIML) was used to estimate missing data. First, descriptive and correlation analyses were conducted. Second, a sequential mediation model was tested linking Wave 1 family income to Wave 3 adolescent marijuana use through Wave 1 family economic pressure, Wave 2 maternal internalizing symptoms, Wave 2 maternal hostility toward partner, and Wave 3 maternal hostility toward adolescent (b1->b2->b3->b4->b5, Figure S1). Third, models estimated the moderating effects of components in Wave 3 adolescent executive function (i.e., working memory and shifting task performance tasks) on the association between Wave 3 maternal hostility toward adolescent and Wave 3 adolescents' likelihood of marijuana use (c1 influence on b5, Figure S1). Each executive function component (i.e., working memory, reaction time in shifting task's stay and switch trials) was tested in separate models due to high correlations among some components. Simple slopes were estimated by probing at low (1 *SD* below mean) and high (1 *SD* above mean) levels of a moderator (Aiken & West, 1991). Fourth, conditional indirect effects were estimated to determine whether the full pathway from Wave 1 family income to Wave 3 adolescents' likelihood of marijuana use varied by Wave 3 adolescent executive function (c1 influence on b1-> b2->b3->b4->b5, Figure S1). Post-hoc power analyses using Monte Carlo simulations (1,000 replications) in Mplus 8.3 indicated that the sample had sufficient power (.80) to detect small (.25)

to moderate (.33) effect sizes in the mediation model. Two sets of sensitivity analyses were also conducted to examine the robustness of the study findings by considering the potential trade-off between accuracy and reaction time in the shifting task performance task (see supplemental analysis plan and Figure S3).

Results

Please see supplemental results for descriptive analyses of study variables (Table 1).

Pathway from Family Income to Adolescents' Likelihood of Lifetime Marijuana Use

Figure S4 shows that all paths from Wave 1 family income to adolescents' likelihood of lifetime marijuana use at Wave 3 were significant in the expected direction. However, the indirect effect of Wave 1 family income to adolescents' likelihood of marijuana use at Wave 3 through Wave 1 family economic pressure to Wave 2 internalizing symptoms to Wave 2 maternal hostility toward partner to Wave 3 maternal hostility toward adolescent was marginally significant (B = -.002, p = .064). This marginal result suggests the strength of the indirect pathway may improve from the influence of certain resilience factors on the mediational process. To examine this possibility, the current study tested whether adolescent executive function may moderate the effects along this pathway.

Moderating Role of Adolescent's Executive Function

Adolescents' working memory at Wave 3 significantly moderated the association between Wave 3 maternal hostility toward adolescent and Wave 3 adolescents' likelihood of marijuana use (Figure 1 and Table 2). Simple slope analysis (Figure 2) shows that when adolescents had lower working memory, higher maternal hostility was associated with a higher likelihood of lifetime marijuana

use among Mexican-origin adolescents (b = .744, SE = .188, p < .001). However, maternal hostility was not associated with adolescents' likelihood of marijuana use among those with higher levels of working memory (b = .180, SE = .175, p = .303). Adolescents' reaction times in stay and switch trials at Wave 3 significantly moderated the association between Wave 3 maternal hostility toward adolescent and Wave 3 adolescents' likelihood of marijuana use (Figure 1 and Table 2). Simple slope analysis shows that higher maternal hostility was significantly associated with a higher likelihood of lifetime marijuana use among Mexican-origin adolescents who have shorter reaction times in switch trials (b = .718, SE = .172, p < .001, but not for adolescents with longer reaction times (b = .073, SE = .186, p = .696) (Figure 3). A similar pattern was observed for stay trials, where maternal hostility was only associated with higher adolescent marijuana use among adolescents with shorter reaction time in stay trials (b = .720, SE = .176, p < .001), whereas no significant association was found for those with longer reaction times (b = .072, SE = .175, p = .680) (Figure S5).

Conditional Indirect Effect from Family Income to Adolescent's Marijuana Use

Table S1 presents the conditional indirect effect from Wave 1 family income to adolescents' likelihood of marijuana use at Wave 3 at different levels of adolescents' working memory and reaction times in stay and switch trials at Wave 3. Results demonstrated that the indirect effect from Wave 1 low family income relating to adolescents' likelihood of marijuana use at Wave 3 was only significant when Mexican-origin adolescents had lower/mean levels of working memory or

shorter reaction times in stay or switch trials[1].

Sensitivity Analysis

Supplemental sensitivity analyses demonstrated the protective role of longer reaction time in preventing marijuana use from maternal hostility only among youth with low accuracy in shifting tasks.

Discussion

This study contributes by identifying potential resilience factors that buffer the detrimental links from family income to adverse adolescent developmental outcomes within the family stress model (Masarik & Conger, 2017). Drawing on longitudinal data from Mexican-origin families, the current study revealed a pathway linking family income to adolescents' lifetime marijuana use through mothers' perceptions of economic pressure, their internalizing symptoms, hostility toward partner, and hostility toward their adolescent children. Moreover, the study highlights adolescent executive function, especially working memory and shifting task performance, as key resilience factors that can buffer this risk pathway.

Main Pathways of Family Stress Model

The findings provide support for the FSM and its applicability to Mexican-origin families (Martin et al., 2019; Masarik & Conger, 2017), by demonstrating that stressors external to the

[1]Additional analyses were conducted by controlling for baseline maternal internalizing symptoms, hostility toward partner, and hostility toward adolescent. Results show that all previously significant individual pathways remained significant, but the conditional indirect effect was no longer significant. Given the study's focus on resilience factors within the family system, the final model retained the original specification without baseline controls.

marital relationship (i.e., low income and economic pressure) can contribute to a member of the couple's psychological distress and increased hostility between partners. These disruptions in interparental functioning may, in turn, culminate in a series of detrimental and hostile mother-child interactions, such as shouting, yelling, and getting into a fight, which can result in diminished openness in mother–child communication, and reductions in adolescents' own subjective experiences of maternal support (Criss et al., 2015). Ultimately, these cascading processes could increase the likelihood of adolescent marijuana use in late adolescence, potentially as a form of maladaptive coping or self-medication to family dysfunction. Thus, adolescents' marijuana use can be a critical problem as it may go unnoticed or unaddressed within the family environment.

As Mexican-origin adolescents, as a group, already face disparities in their health outcomes and substance use rates (Delva et al., 2005), identifying how family economic hardship can explain this substance use disparity is critical for building and developing appropriate interventions to mitigate this risk. However, it is worth noting that the indirect effect in the full mediation model was only marginally significant, which is not consistent with previous findings using the FSM framework (Delgado et al., 2013; Taylor et al., 2024). This suggests that certain pathways may be conditional on the presence of potential moderating processes or resilience factors. This possibility was further explored through moderation analyses in this study.

Executive Function as a Resilience Factor in the Family Stress Model

Critically, the current study tested adolescent executive function, specifically working memory and shifting task performance (including baseline attention and re-oriented attention), as

protective factors within the family stress model. Consistent with existing literature, better working memory, as indicated by longer digit span, buffered the link between maternal hostility toward adolescent and youth marijuana use (Khurana et al., 2013). Adolescents with stronger working memory may retain alternative responses to maternal hostility, reducing reliance on maladaptive coping behaviors like marijuana use. This aligns with broader evidence that executive function training can mitigate substance use risk, underscoring its relevance for interventions in high-stress contexts (Khemiri et al., 2019).

For shifting task performance, the results suggested that Mexican-origin adolescents' longer, rather than shorter reaction times in both stay and switch trials, protected against the impact of maternal hostility on lifetime marijuana use. Contrary to conventional interpretations, longer reaction times in shifting tasks emerged as protective, challenging assumptions that faster responses universally signify better performance in cognitive tasks. In this sample of bilingual youth from low-income Mexican immigrant families, longer reaction times may reflect adaptive deliberation rather than inefficiency. When experiencing systemic stressors, slower responses could indicate reduced impulsivity, allowing adolescents to engage in controlled processing and disengagement from hostile interactions. This aligns with critiques of using reaction time as a standalone metric (Draheim et al., 2019), highlighting the need to contextualize executive function measures within cultural and environmental demands. For bilingual youth navigating acculturative and economic stressors, slower responses may represent strategic trade-offs to prioritize accuracy or emotional regulation over speed (Draheim et al., 2016), which may protect them from the negative impact of maternal hostility on marijuana use. The three-way interaction

further clarifies this mechanism: the protective effect of longer reaction times was significant only for adolescents with lower accuracy. This suggests that youth with poorer task performance may deliberately slow down to compensate, mitigating errors and impulsive coping. Such findings emphasize the importance of reconceptualizing executive function assessments to account for compensatory strategies in marginalized populations, where conventional metrics may overlook culturally adaptive processes.

Taken as a whole, a strength of the current study is its examination of executive function as a novel adolescent individual-level buffering factor, especially working memory and shifting task performance, which is relatively understudied and rarely applied in the context of the FSM. The findings also highlight the practical potential of working memory and shifting task performance (including baseline attention and re-oriented attention) as modifiable targets for attenuating the impact of family stress process on adolescent marijuana use. This is consistent with evidence from intervention studies suggesting that improvements in working memory and shifting task performance can lead to reductions in substance use behaviors (Khemiri et al., 2019). It is also important to highlight the fact that there was only a small proportion of adolescents who reported marijuana use in their lifetime, so much so that the analysis had to be restructured to a logit regression (i.e., dichotomize the outcome variable). This finding is consistent with previous research that has documented similar rates of "non-users" for cannabis use in a Mexican-origin sample (~74%) (Cruz et al., 2018). Notably, the fact that only 34.6% of adolescents in the sample reported *lifetime* marijuana use is encouraging and may be instrumental in challenging stereotypes about substance use in Mexican-origin youth. It also indicates that many

Mexican-origin adolescents continue to succeed and thrive despite their disadvantaged financial situation, which suggests the presence of resilience buffers and protective factors that are present and salient in adolescents' lives. It is important for future work to continue investigating other resilient factors among Mexican-origin samples when examining the FSM. Also, the current study did not control for baseline levels of maternal internalizing symptoms, hostility toward partner, and hostility toward adolescents. When additional analyses were conducted to control for such baseline levels, results revealed that all previously significant individual pathways remained significant, providing further support for the family stress model. However, the conditional indirect effect was no longer significant after accounting for these baseline variables. This attenuation may be due to a reduction in available variance that could be explained by executive function as a resilience factor. Given that the central aim of this study was to investigate executive function as a protective factor within the family stress model, the final model presented in this study is without adjusting for baseline maternal internalizing symptoms and hostility toward adolescent.

Implications

Several implications can be drawn from the present findings. By investigating possible resilience processes in a Mexican-origin community, which is highly susceptible to low family income, the study expands the FSM to show how adolescents' assets (executive function) can strengthen resilience in times of economic adversity, an area that is relatively less studied.

Family scholars can also contribute to more effective solutions that nurture family cohesion and positive developmental outcomes for adolescents from disadvantaged communities.

Specifically, policies aimed at alleviating economic hardship may be implemented to prevent the escalation of familial economic pressure to disrupted familial dynamics and to adolescents' substance use. Those policies can be particularly critical for youth from Mexican-origin families considering that the systematic barriers they may encounter may hinder their access to health services, limiting the resources and support available to prevent and treat health problems related to substance use.

This study's novel findings on executive function in adolescent coping highlight the importance of integrating individual-level resilience-building strategies into interventions for adolescents facing economic hardship and family stress, alongside family-based efforts to reduce parental stress and hostility. Strengthening executive functions may help mitigate reliance on maladaptive coping behaviors like marijuana use by enhancing adolescents' ability to focus on their own activities or goals and shift attention away from stressors. For example, school-based programs incorporating executive function training and curriculum-integrated metacognition modules (e.g., planning, self-monitoring) may help buffer against the cumulative toll of chronic stress (Camuñas et al., 2022), particularly in economically disadvantaged communities where systemic inequities exacerbate familial strain. Future research should examine how embedding executive function training into existing prevention programs may enhance their effectiveness in reducing adolescent substance use, especially in high-risk populations.

Strengths and Limitations

The focus on Mexican-origin families is a key strength of the current study, as this population represents individuals who are particularly vulnerable to economic hardship and its

associated risks. In addition, given that many studies on the FSM are cross-sectional in nature, it is a strength of the present work that a longitudinal sample was used, allowing researchers to make a stronger causal inference from family economic adversity to adolescent substance use and revealed how the associations unfolded over time.

Some limitations of the current work need to be highlighted. First, most recruited families resided in a metropolitan city in central Texas with parents being first-generation immigrants and adolescents being born in or coming to the U.S. at a young age. Thus, findings may not generalize to other Mexican-origin families in different contexts, such as non-immigrant families or families from other geographical spaces. Studies are needed to extend the present findings for the FSM among a more heterogenous Mexican-origin sample. Next, the study solely examined mother-adolescent dynamics, as mothers typically transmit culture in Latino families (Padilla, 2006), and did not examine fathers' potential influence. Given the mixed findings concerning paternal familism and the FSM (Martin et al., 2019; White et al., 2015), future research should include both parents to fully understand the family process in the link between economic pressure and adolescent substance use. Third, several key variables were measured at the same wave (e.g., family income and economic pressure at Wave 1, and maternal internalizing symptoms and partner hostility at Wave 2), which limits the possibility to draw directional inferences for certain pathways. Moreover, the study also did not control for maternal internalizing symptoms and hostility at earlier waves. Future work with a focus on elucidating longitudinal changes in maternal internalizing symptoms and hostility toward partner may be necessary to expand on the current findings such as by using cross-lagged panel models to test

for potential bidirectionality of study constructs. Lastly, due to small sample size in marijuana use categories, marijuana use was dichotomized as use versus no-use. Future studies with larger samples are necessary; perhaps by further exploring varying intensities of marijuana use, which would help validate and extend the present findings.

Conclusion

The current study contributes to the FSM literature by investigating how potential resilience buffers, including adolescents' working memory and shifting task performance, can protect adolescents against substance use and support Mexican-origin families in thriving and succeeding despite economic challenges. In addition to extending theoretical conceptions of the FSM by examining potential resilience factors, the use of a longitudinal sample in the present investigation also overcomes past criticisms of the cross-sectional nature of many existing FSM studies. The utilization of a specific sample of Mexican-origin families suggests the applicability of FSM to vulnerable populations most susceptible to economic hardship. Importantly, policies aimed at alleviating economic hardship and developing appropriate intervention that capitalizes on adolescent executive function may aid Mexican-origin families in overcoming their economic disadvantage and excelling in life.

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